

Tyler Chinsky

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Computational biologist with an M.S. in Bioinformatics and six years of experience in genomics, transcriptomics, and machine learning (ML), with a focus on oncology and cancer research spanning clinical trial biomarker discovery, next-generation sequencing (NGS) analysis, CAR T-cell genomics, and high-throughput bioinformatics pipeline development across cloud and high-performance computing (HPC) environments.

TECHNICAL SKILLS

Genomic & Bioinformatic Analysis: Next-generation sequencing (NGS), bulk/single-cell/spatial RNA-seq, WGS/WES/WXS, multiome, HiC-seq/ChIP-seq/ATAC-seq, GSEA/GSVA, clinical biomarker discovery, immunogenomics, immuno-oncology, structural & functional protein analysis

Programming & Pipelines: Python, R, WDL, Nextflow, Docker/Apptainer (Docker & Kubernetes certified), cloud computing, high-performance computing (HPC), machine learning (ML), agentic AI, C++, Java, Scala

Wet Lab: Next-generation sequencing (NGS) library prep (Illumina NextSeq/NovaSeq), DNA/RNA extraction, PCR, protein purification, multiplex immunofluorescence imaging

EXPERIENCE

Computational Biologist | Genomics and Bioinformatics Core, University at Buffalo 2024 - Present

- Lead analysis projects for internal and external customers spanning spatial biology, single-cell, bulk, and multiome sequencing
- Lead development and integration of scalable, containerized bioinformatics workflows (Docker/Apptainer/Nextflow) for standard core analyses
- Lead integration of agentic AI and ML tooling into core workflows to improve analyst efficiency and reduce project turnaround time
- Contributed to multiple peer-reviewed publications arising from core-supported research projects

Computational Biologist (Consultant) | D2Dx Consulting 2024

- Analyzed clinical and molecular data to identify candidate biomarkers supporting oncology clinical trial design
- Applied GSVA/GSEA methods to evaluate gene expression signatures against clinical and molecular annotations

Computational Biologist I | Medical Oncology Department, Dana-Farber Cancer Institute 2022 - 2024

- Built and maintained data processing and analysis pipelines for high-throughput clinical DNA and RNA sequencing, with an emphasis on immune profiling
- Developed new algorithms to meet emerging clinical analysis needs and integrated tools/databases into existing high-throughput pipelines
- Extracted and analyzed spatial data from multiplex immunofluorescence imaging in support of collaborative, co-authored oncology studies

Affiliate Scientist | Cancer Program, Broad Institute of MIT and Harvard 2022 - 2024

- Leveraged Broad Institute computational resources and datasets to support oncology research conducted at Dana-Farber Cancer Institute

Computational Biologist | Sinha Lab & Seiffert Lab, University at Buffalo 2020 - 2023

Pemphigus vulgaris HLA binding site identification and genomic expression analysis using NGS

- Built and executed pipelines in Python, R, and Bash to analyze microarray, RNA-seq, scRNA-seq, and genomic data, comparing expression profiles across disease and control cohorts
- Designed experiments to functionally validate NGS-derived signatures, focusing on transcriptome profiling of immunological pathways
- Contributed to collaborative, co-authored publications

Associate Computational Biologist | Cui Lab, Rochester Institute of Technology 2015 - 2016

Chromatin organization associated with DNA binding of tumor suppressor TP53

- Developed ChIP-seq analysis pipelines to identify genomic regions inhibiting TP53 DNA-binding sites
- Analyzed gene expression data for pathway activation/inhibition at specific TP53 binding regions

EDUCATION

M.S., Bioinformatics, Johns Hopkins University | Baltimore, MD

May 2025

B.S., Bioinformatics and Computational Biology, University at Buffalo | Buffalo, NY

Feb 2022

PUBLICATIONS

Leick, M.B., Sun, B., Birocchi, F., ... **Chinsky, T.M.**, ... Maus, S.V. (2026). Genomic correlates of clinical CAR T cell activity. *Science Immunology*, 11(121), eaef4134. <https://doi.org/10.1126/sciimmunol.aef4134>

Hamid, A.A., Vergara, M., Park, J., **Chinsky, T.M.**, ... Sweeney, C.J. (2024). Whole exome correlates in metastatic hormone sensitive prostate cancer (mHSPC): results from E3805 CHAARTED. *J Clin Oncol*, 42, 5088. https://doi.org/10.1200/JCO.2024.42.16_suppl.5088

Hamid, A.A., **Chinsky, T.M.**, Vergara, M., ... Sweeney, C.J. (2023). Landscape and impact of germline pathogenic variants in metastatic hormone sensitive prostate cancer (mHSPC): ancillary study of E3805 CHAARTED. *J Clin Oncol*, 41(16_suppl), 5082. https://doi.org/10.1200/JCO.2023.41.16_suppl.5082

Bowitch, A., **Chinsky, T.M.**, Yu, M.C., Ferkey, D.M. (2022). The *C. elegans* OCTR-1 and human alpha-2A adrenergic receptors are methylated within the third intracellular loop by human PRMT5 in vitro. *microPublication Biology*. <https://doi.org/10.17912/micropub.biology.000546>.

Chinsky, T.M., Chen, R., Holan, S., Seiffert, K.R., Sinha, A.A. (2022). Investigation of transcriptionally regulated differential gene expression as a potential protection signature in the pemphigus vulgaris disease model. SUNY Buffalo Jacobs School of Medicine Research Symposium [Poster/Abstract].